

Rear Bike Light — Hardware Specification (ATtiny13A)

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Overview

A rear bicycle light for a local cycling club, built around an ATtiny13A powered directly from a single protected Li-ion cell. The hardware is frozen at schematic revision circuit2; firmware has not been started.

Design goals, in priority order:

- Easy for any club member to build, repair and adapt (common parts, hand-solderable, one resistor sets LED current).
- Reliable and safe without relying on firmware alone (protected cells, ballast resistor, BOD, watchdog).
- Sleep current under 1 μA , so the light can sit in a bag for months.
- Programmable in-circuit with a SOIC-8 clip, no desoldering.
- Usable both as a complete new board and as a retrofit into existing cheap blinker housings.

Power and charging

VCC is the battery itself: there is no regulator, no on-board protection and no power switch. Everything below assumes a single Li-ion/LiPo cell of 300–400 mAh.

Battery (BT1). The cell must have its own protection board (PCM). Club members confirmed their cells already do, so the board carries none. If unprotected cells (e.g. bare 10440) ever need support, add a single-chip protector with integrated MOSFET such as XB5352A or XB5353A. Add a text note on the schematic: "BT1 — protected cells only".

Charger (U2, TP4057, SOT-23-6). Pinout: 1 CHRG, 2 GND, 3 BAT, 4 VCC, 5 STDBY, 6 PROG.

- R2 on PROG sets the charge current. Default 4.7 k Ω $\approx$ 175 mA, about 0.5C for 350 mAh. 2 k Ω $\approx$ 0.4 A is for larger cells only. The Rprog formula differs between TP4057 vendors, so verify against the datasheet of the part actually bought.

- Higher current also heats U2: at 0.4 A it dissipates about 0.55 W and enters thermal regulation.
- CHRG (D2) and STDBY (D1) share R3 = 1 k Ω from VBUS; only one is lit at a time. Both may blink with no battery, which is normal.
- The load sits in parallel with the battery. If the light is on while charging, the charger may never reach its C/10 termination threshold. There is no charge-detect input to the MCU (PB1 is free for it later).

USB (J2). USB-C power-only 6-pin receptacle (KiCad symbol

USB_C_Receptacle_PowerOnly_6P). CC1 (A5) and CC2 (B5) each have 5.1 k Ω to GND (R5, R4); without them C-to-C cables will not supply power. Connect SHIELD to GND (still open in circuit2).

Decoupling.

Ref	Value	Location	Purpose
C2	10 μ F	VBUS, at U2	Input bulk; electrolytic or tantalum preferred, its ESR damps hot-plug ringing
C3	10 μ F	BAT, at U2 pin 3	Charger output stability and battery rail bulk
C1	100 nF	ATtiny pins 8/4	Local MCU decoupling

MCU pinout (U1, ATtiny13A, SOIC-8)

The LED stays on PB0 because only PB0/PB1 have hardware PWM (OC0A/OC0B); the button moved off the ISP lines.

Pin	Port	Alt. function	Connection	Notes
1	PB5	RESET, ADC0	R1 10 k Ω to VCC	Keep as RESET for ISP; never set RSTDISBL
2	PB3	ADC3, PCINT3	SW1 to GND	Internal pull-up; pin-change wake from power-down
3	PB4	ADC2	Divider midpoint (R11/R7) + C4 10 nF	Battery voltage input

Pin	Port	Alt. function	Connection	Notes
4	GND	—	GND	
5	PB0	OC0A, MOSI	R8 100 Ω to Q1 gate; R10 100 k Ω gate to GND	LED PWM
6	PB1	OC0B, MISO	Not connected	Bring out to a pad (future USB detect, NTC)
7	PB2	ADC1, SCK	R11 1 M Ω to divider top	Divider enable: HIGH to measure, LOW otherwise
8	VCC	—	Battery +	C1 100 nF close to pin

LED driver

The LED is switched by a low-side MOSFET with a ballast resistor, and firmware PWM sets the average current. Topology: VCC $\rightarrow$ R9 $\rightarrow$ D3 (anode $\rightarrow$ cathode) $\rightarrow$ Q1 drain; Q1 source $\rightarrow$ GND.

- **Q1: AO3400A** (SOT-23), or SI2302 / AO3416. $R_{ds(on)}$ is about 30–50 m Ω at V_{gs} 2.5 V, so it is fully on at any working battery voltage and stays cool.
- **Gate:** R8 = 100 Ω in series from PB0; R10 = 100 k Ω gate to GND keeps Q1 off during reset, programming, power-up and after a watchdog reset.
- **PWM frequency:** a few kHz or higher, to avoid visible stroboscopic flicker.
- **R9** is the only part a builder changes to match their LED. Default 10 Ω suits 0.5–1 W red LEDs: peak about 190 mA at 4.2 V, 110 mA at 3.4 V, up to 0.36 W at 100% duty.
- **Footprint:** use 2512 pads (2010 also fits); 1206 is too small. Optionally provide two R9 footprints in series at opposite board edges, the second fitted with 0 Ω by default, so heat can be split.

The ballast resistor is mandatory. Without it, peak current is limited only by milliohms of battery, MOSFET and wiring resistance: 2–4 A on a red LED, which no PWM frequency makes safe.

Sizing R9

Pick V_{full} , the battery voltage down to which full brightness is held. Below it firmware runs 100% duty and brightness falls until cutoff.

$$R_9 = \frac{V_{full} - V_f}{I_{target}}$$

Firmware scales duty with battery voltage: $D = I_{target} \times R_9 / (V_{bat} - V_f)$, clamped to 100%. Resistor heat does not depend on duty: $P = I_{target} \times (V_{bat} - V_f)$, worst at 4.2 V. Efficiency is about V_f / V_{bat} , roughly 60% for a red LED, the same as any linear driver.

Examples for a red LED ($V_f \approx 2.3$ V), $V_{full} = 3.4$ V:

Average current	R9 total	As two resistors	Peak at 4.2 V	Total heat at 4.2 V
150 mA	7.3 Ω	2 $\times$ 3.6 Ω	0.26 A	0.29 W
250 mA	4.4 Ω	2 $\times$ 2.2 Ω	0.43 A	0.48 W
350 mA	3.1 Ω	2 $\times$ 1.6 Ω	0.6 A	0.67 W

Check the LED's pulse current rating: 0.6 A peak is borderline for many 1 W red LEDs. If it is too high, raise V_{full} (e.g. 3.5 V), accepting earlier dimming.

The 10 Ω default will destroy 5 mm LEDs (20 mA). Document an R9 table per LED type for builders (see Retrofit notes), and make firmware start at a low brightness on first boot.

Battery voltage measurement

Battery voltage is read through a switched resistive divider into ADC2, against the internal 1.1 V reference. The ATtiny13A cannot measure its own VCC: its ADC mux has no bandgap input channel (unlike ATtiny25/45/85).

Circuit. PB2 $\rightarrow$ R11 1 M Ω $\rightarrow$ PB4 (ADC2) $\rightarrow$ R7 270 k Ω $\rightarrow$ GND, with C4 10 nF from PB4 to GND.

- Ratio 270 / 1270 = 0.2126, so 4.2 V reads as 0.893 V.
- The 1.1 V reference varies $\pm 10\%$ per chip. Even at 1.0 V, full scale is 4.70 V, so a charging cell never saturates the ADC.
- Resolution is about 5 mV of battery voltage per LSB. Use 1% resistors.
- C4 supplies the ADC sample charge, since the divider source impedance (≈ 212 k Ω) is far above the recommended 10 k Ω .

Switching. Drive PB2 HIGH to measure. The time constant is 212 k Ω $\times$ 10 nF $\approx$ 2.1 ms, so wait at least 10 ms before converting. At all other times, including sleep, drive PB2 output LOW (not high-Z), so both ends of the divider sit at GND and draw no current. Disable the

PB4 digital input buffer (DIDR0, ADC2D), because the node sits at an intermediate voltage while measuring.

Calibration. The reference tolerance makes per-unit calibration mandatory: measure the battery once with a multimeter and store a correction factor in EEPROM, via a service mode.

When to measure. For cutoff decisions, measure with the LED on (under load), synchronised to the PWM on-phase or with a short 100% pulse. A small, cold cell can sag by a few tenths of a volt under load.

Fuses, BOD, watchdog and sleep

High fuse = 0xFB (BOD at 2.7 V); the low fuse (clock) is decided with the firmware.

Brown-out detector. BODLEVEL options: 0xFF disabled, 0xFD $\approx$ 1.8 V, 0xFB $\approx$ 2.7 V (each ± 0.2 V); 4.3 V is unusable on Li-ion. BOD is not battery protection: it only holds the MCU in reset and does not stop the drain. Its jobs here:

- Guarantee correct operation at 9.6 MHz, which requires at least 2.7 V.
- Protect EEPROM (calibration, brightness limit) from corruption at low VCC.
- Turn it off during sleep in software (BODS/BODSE in MCUCR), so it costs nothing in power-down.

Never program RSTDISBL (high fuse bit 0): it disables ISP and only a high-voltage programmer can recover the chip.

Watchdog. Enable in software only; never program WDTON, or it cannot be disabled before sleep.

- Timeout 2 s or 4 s (available steps: 16 ms to 8 s).
- Reset it only in the main loop, never from an ISR.
- At the very start of main, clear WDRF in MCUSR and disable the WDT. After a WDT reset it stays enabled at 16 ms and causes a reset loop otherwise.
- After any reset all pins are high-Z and R10 turns the LED off, so the safe state needs no code.

Sleep (target < 1 μ A). Power-down with WDT off, BOD off via BODS, ADC off, PB2 driven LOW, PB4 digital input disabled, PB3 pull-up on for pin-change wake.

Programming (ISP) and safety rules

Program in-circuit with a SOIC-8 test clip, with the battery connected and the programmer not supplying power.

- **Never feed 5 V from the programmer.** VCC is wired straight to the Li-ion cell. On USBasp, remove the target-power jumper; keep a club cable with the VCC wire cut and labelled. Programmers that only sense target voltage (AVRISP mkII, Atmel-ICE) are safe as is.
- A 3.3 V programmer is preferable for signal levels; a 5 V one usually works through the input clamp diodes.
- Line loading is harmless: MOSI (PB0) drives the gate, so the LED may flicker during programming; SCK (PB2) sees the 1.27 M Ω divider; MISO (PB1) is free; RESET has R1 10 k Ω pull-up.
- Never program RSTDISBL or WDTON (see Fuses).
- No series diode in the supply path for development: its voltage drop would make development and production boards behave differently.

Bill of materials

Values as in schematic circuit²; packages not fixed in the schematic are suggestions for hand soldering.

Ref	Value	Package	Notes
U1	ATTiny13A	SOIC-8	
U2	TP4057	SOT-23-6	Verify vendor Rprog formula
Q1	AO3400A	SOT-23	Alt: SI2302, AO3416
J2	USB-C receptacle, power-only 6P	SMD/THT	SHIELD to GND
BT1	Li-ion/LiPo 3.7 V, 300–400 mAh	—	Built-in protection required; check connector polarity
D3	Red LED	1 W star (default)	Builder's choice; sets R9
D1, D2	Indicator LEDs	0805 suggested	STDBY / CHRG; colours open

Ref	Value	Package	Notes
SW1	Momentary tactile switch	—	Not latching
R1	10 k Ω	0805	RESET pull-up
R2	4.7 k Ω	0805	Charge current $\approx$ 175 mA
R3	1 k Ω	0805	Indicator LEDs
R4, R5	5.1 k Ω	0805	USB-C CC pull-downs
R7	270 k Ω 1%	0805	Divider bottom
R8	100 Ω	0805	Gate series
R9	10 Ω (default)	2512	LED ballast; per-build value
R10	100 k Ω	0805	Gate pull-down
R11	1 M Ω 1%	0805	Divider top
C1	100 nF	0805	MCU decoupling
C2	10 μ F	Electrolytic/tantalum	VBUS input
C3	10 μ F	0805/1206 or tantalum	BAT rail
C4	10 nF	0805	ADC input

Firmware requirements from the hardware

These are constraints the hardware imposes; modes, UI and code structure are still open.

1. **Startup:** first clear WDRF and disable the WDT; then check BORF in MCUSR and treat a brown-out reset as a low-battery event (reduce brightness or shut down), to avoid a flicker loop on a sagging cold cell.
2. **LED PWM:** Timer0 fast PWM on OC0A (PB0), at least a few kHz.
3. **Current control:** $D = I_{\text{target}} \times R9 / (V_{\text{bat}} - V_f)$, clamped to 100%. R9 and V_f are per-build parameters (compile-time or EEPROM).
4. **Brightness limit:** a user-set maximum level stored in EEPROM via a button service menu; low default on first boot.

5. **Battery:** switched divider as described; measure under load; 3–4 charge levels with hysteresis; cutoff at about 3.2–3.3 V under load, then LED off and power-down.
6. **Calibration:** service mode that stores an ADC reference correction factor in EEPROM.
7. **Button:** PB3 with internal pull-up, pin-change interrupt for wake, debounce in software.
8. **Watchdog:** 2 s or 4 s, reset only in the main loop, disabled before sleep.
9. **Sleep:** power-down with WDT, BOD (BODS), ADC off, PB2 LOW, PB4 digital input disabled.
10. **Charging:** no charge-detect input; the light may be left on while charging.

Rejected alternatives

Recorded so they are not proposed again without new information.

Alternative	Why rejected
Measure VCC via internal bandgap as ADC input	ATtiny13A has no bandgap input channel; needs an external divider
AMC7135 constant-current driver	With a red LED it dissipates 0.5–0.65 W in one SOT-89; fixed 350 mA
MOSFET direct drive, no ballast resistor	Uncontrolled 2–4 A peaks; no PWM frequency makes it safe
Capacitor across the LED	Does not limit current; adds inrush on the MOSFET and battery
Software buck on the ATtiny	~200 μ H inductor at 37.5 kHz, slow loop, LED dies if firmware hangs
LED current sense replacing the divider	V_f unknown per LED and temperature; battery voltage accuracy matters more for safety
Series Schottky in the supply during development	Variable drop makes dev and production boards differ; protect at the programmer instead
TP4056 module with DW01 protection	Replaced by bare TP4057 plus protected cells

Alternative	Why rejected
Fixed always-on divider	Replaced by GPIO-switched divider: zero sleep current at no extra parts

Future efficiency upgrade, if ever needed: a boost LED driver with two red LEDs in series and a low-FB (≤ 0.2 V) controller.

Retrofit notes and open questions

When reusing the LEDs of an existing blinker, set R9 from the LED type, using the low end of the range when unsure.

LED type	Typical current per LED
5 mm through-hole	20 mA
SMD 3528	20 mA
SMD 5050 (3 chips)	60 mA
"Piranha" / superflux, 4 legs	50–70 mA
SMD 2835	60–150 mA (varies by wattage)
0.5–1 W on star	150–350 mA

- If the blinker still works, measure its original current with its original batteries in the brightest steady mode, and treat that as the upper limit.
- Coin-cell blinkers often have no LED resistors and rely on the cell's 10–20 Ω internal resistance; on Li-ion without the right R9 they burn out.
- Faulty unit: test each LED with a multimeter diode range or a current-limited supply (5–10 mA).
- Parallel LEDs: give each its own resistor; with a shared one the lowest- V_f LED takes most of the current.
- Thermal test: 15 min in the brightest steady mode, housing closed; it must stay comfortably warm, not hot.
- Cut every trace to the old circuit (including any epoxy-blob MCU).
- The button must be momentary; replace latching buttons or slide switches.
- A kept USB-C socket needs 5.1 k Ω on CC1/CC2; many cheap blinkers omit them.

Open questions

- ☐ Light modes and their currents (longest steady mode drives thermal design)
- ☐ Connect J2 SHIELD to GND
- ☐ Add "protected cells only" note to the schematic
- ☐ Bring PB1 out to a pad
- ☐ Second R9 footprint (0 Ω default) at the opposite board edge
- ☐ Indicator LED colours (D1, D2)
- ☐ Clock frequency and low fuse
- ☐ Exact Rprog formula for the chosen TP4057 vendor
- ☐ R9-per-LED table on silkscreen or in the build guide
- ☐ How Vf is configured per build (compile-time vs EEPROM)